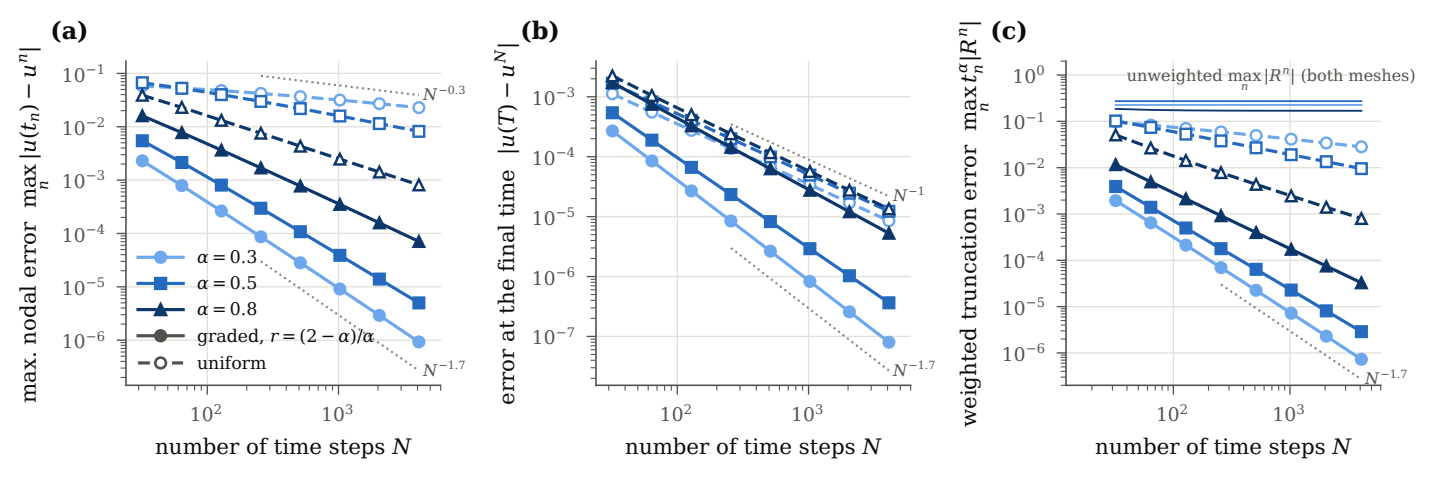
(a)
max. nodal error $\max_n |u(t_n) - u^n|$
number of time steps N
$N^{-0.3}$
$N^{-1.7}$
$\alpha = 0.3$
$\alpha = 0.5$
$\alpha = 0.8$
graded, $r = (2-\alpha)/\alpha$
uniform
(b)
error at the final time $|u(T) - u^N|$
number of time steps N
N^{-1}
$N^{-1.7}$
(c)
weighted truncation error $\max_n t_n^{\alpha} |R^n|$
unweighted $\max_n |R^n|$ (both meshes)
number of time steps N
$N^{-1.7}$